Day 6: Generative Modeling & DDPM

Welcome

- **Generative Modeling & DDPM** — The variational view — from VAEs to diffusion
- 3 hours lecture + practical
- Slides, notes, and code on the course site

Outline

Deep Generative Modeling

The Generative Goal

- Map a **simple** distribution (Gaussian noise) to a **complex** one (data)
- Sample $x \sim p_{\text{data}}$ we can only access through examples
- Diffusion: don't jump noise→data in one step — move in many small steps
- Forward = corrupt data into noise; reverse = learn to undo it
- Source: Principles of Diffusion Models, Ch. 1–2

The Generative Goal — illustration

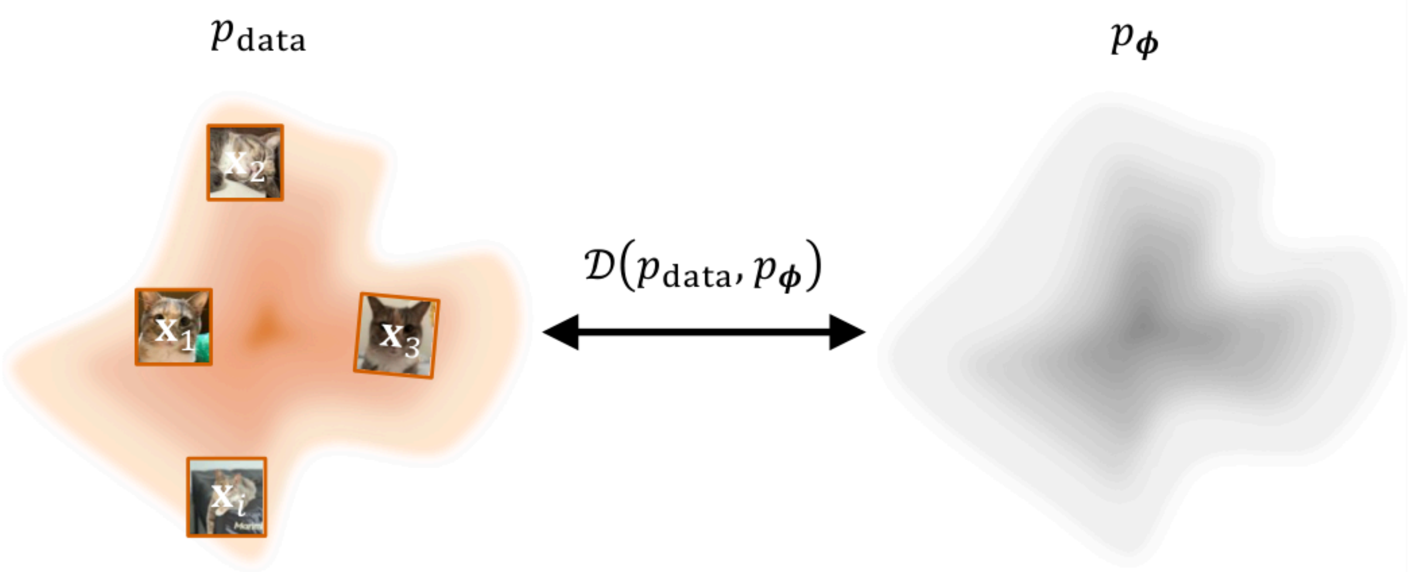
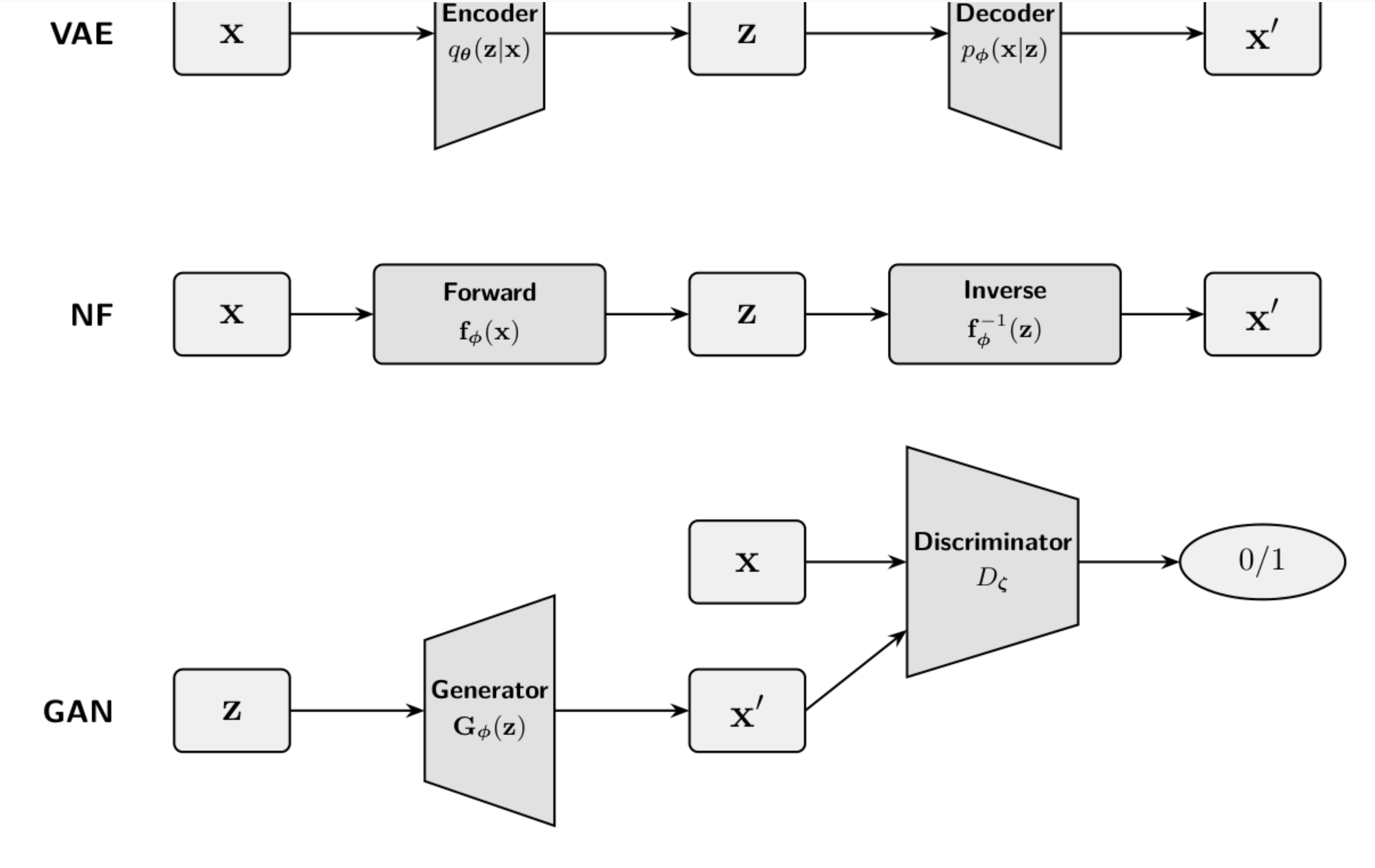


Figure 1.1: Illustration of the target in DGM. Training a DGM is essentially minimizing the discrepancy between the model distribution p_ϕ and the unknown data distribution p_{data}. Since p_{data} is not directly accessible, this discrepancy must be estimated efficiently using a finite set of independent and identically distributed (i.i.d.) samples, $\mathbf{x}_i$, drawn from it.

Taxonomy of Models

- Likelihood-based: VAEs, normalizing flows, autoregressive, diffusion
- Implicit: GANs (no tractable $p(x)$)
- Trade-offs: sample quality vs exact likelihood vs sampling speed
- Diffusion = likelihood-based + high quality, but slow sampling

Taxonomy of Models — illustration



Taxonomy of Models — illustration

Deep Generative Modeling — Taxonomy of Models (source: course materials)

Latent-Variable Models

- Introduce latent z: $p_\theta(x) = \int p_\theta(x|z)p(z)\,\mathrm{d}z$
- Prior $p(z) = N(0, I)$ is easy to sample
- Marginal likelihood is intractable (integral over all z)
- Posterior $p_\theta(z|x)$ also intractable $\rightarrow$ variational inference

Variational Autoencoders

The Evidence Lower Bound

- Encoder $q_\varphi(z|x)$ approximates the true posterior
- $\log p_\theta(x) \geq \mathbb{E}_{q_\varphi}[\log p_\theta(x|z)] - D_{\mathrm{KL}\left(q_\varphi(z|x) \,\|\, p(z)\right)}$
- Reconstruction term + regularization term
- Gap to $\log p(x)$ is exactly $D_{\mathrm{KL}\left(q_\varphi(z|x) \,\|\, p_\theta(z|x)\right)} \geq 0$

The Evidence Lower Bound — illustration

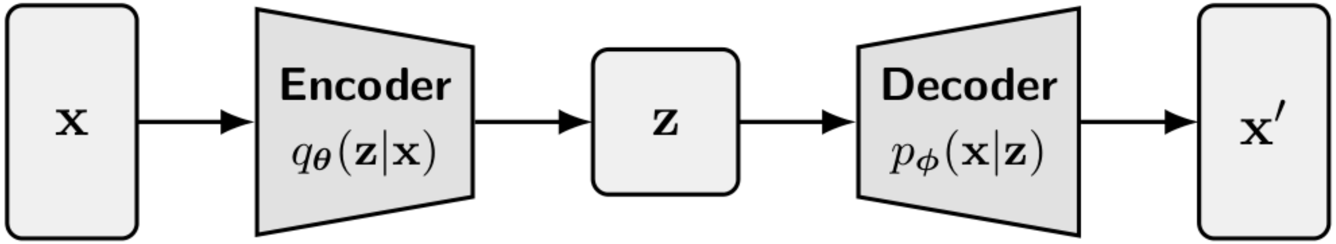


Figure 2.1: Illustration of a VAE. It consists of a stochastic encoder $q_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})$ that maps data $\mathbf{x}$ to a latent variable $\mathbf{z}$, and a decoder $p_{\boldsymbol{\phi}}(\mathbf{x}|\mathbf{z})$ that reconstructs data from the latent.

Source: Created by the authors

Variational Autoencoders — The Evidence Lower Bound (source: course materials)

Reparameterization Trick

- Sample $z = \mu_\varphi(x) + \sigma_\varphi(x) \cdot \varepsilon, \varepsilon \sim N(0, I)$
- Moves randomness off the computation path $\to$ low-variance gradients
- Backprop flows through $\mu_\varphi, \sigma_\varphi$
- DDPM = a **deep hierarchy** of these latents with a fixed encoder

The Forward (Noising) Process

Unified Forward Rule

- $x_t = \alpha_t x_0 + \sigma_t \varepsilon, \varepsilon \sim N(0, I)$
- $p(x_t \mid x_0) = N\left(x_t; \alpha_t x_0, \sigma_t^2 I\right)$
- α_t = signal kept; σ_t = noise added
- Signal-to-noise ratio $\mathrm{SNR}(t) = \frac{\alpha_t^2}{\sigma_t^2}$ decreases with t

Unified Forward Rule — illustration

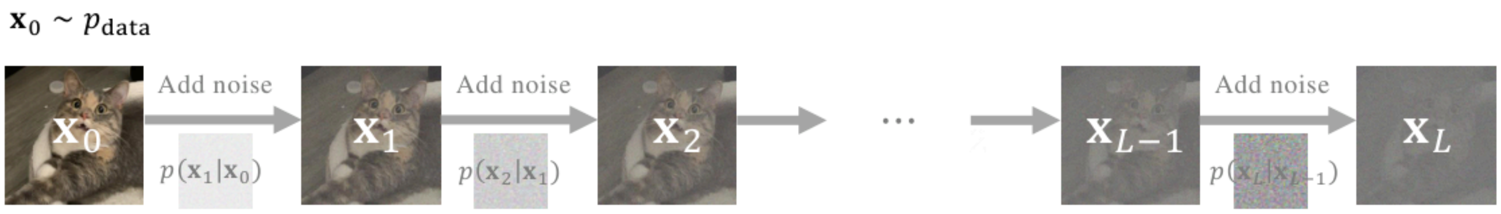


Figure 2.4: Illustration of the DDPM forward process, wherein Gaussian noise is incrementally added to corrupt a data sample into pure noise.

The Forward (Noising) Process — Unified Forward Rule (source: course materials)

Closed-Form Marginal

- Step kernel $q(x_t \mid x_{t-1}) = N\left(\sqrt{1-\beta_t}x_{t-1}, \beta_t I\right)$
- Compose Gaussians: $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon$
- $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$ so $\alpha_t = \sqrt{\bar{\alpha}_t}$
- Sample any t in one shot — no need to simulate the chain

Noise Schedules

- VP (DDPM/cosine): $\alpha_t^2 + \sigma_t^2 = 1$
- VE (EDM): $\alpha_t = 1$, $\sigma_t = t$
- Linear interpolation (flow matching): $\alpha_t = 1 - t$, $\sigma_t = t$
- Interactive — compare schedules on the same image (see notes)

Noise Schedules — illustration

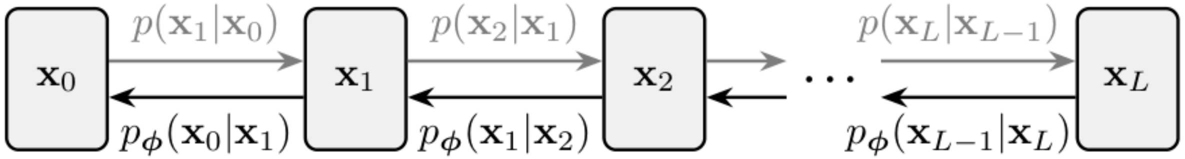


Figure 2.3: Illustration of DDPM. It consists of a fixed forward process (in gray) that gradually adds Gaussian noise to the data, and a learned reverse process that denoises step-by-step to generate new samples.

The Forward (Noising) Process — Noise Schedules (source: course materials)

The Reverse (Denoising) Process

Reverse as Denoising

- Start at $x_T \sim N(0, I)$, walk back to data
- Learn $p_\theta(x_{t-1} \mid x_t)$ — one denoising step
- Oracle reverse kernel $p(x_{t-1}|x_t)$ is intractable...
- ...but $q(x_{t-1} \mid x_t, x_0)$ **is** Gaussian (condition on x_0)

Reverse as Denoising — illustration

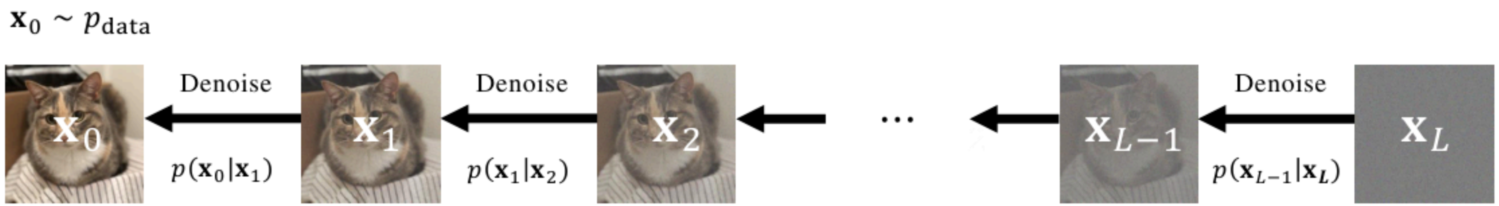


Figure 2.5: Illustration of DDPM reverse (denoising) process. Starting from noise $\mathbf{x}_L \sim p_{\text{prior}}$, the model sequentially samples $\mathbf{x}_{i-1} \sim p(\mathbf{x}_{i-1}|\mathbf{x}_i)$ for $i = L, \ldots, 1$ to obtain a newly generated data $\mathbf{x}_0$. The oracle transition $p(\mathbf{x}_{i-1}|\mathbf{x}_i)$ is unknown; thus, we aim to approximate it.

The Reverse (Denoising) Process — Reverse as Denoising (source: course materials)

The True Posterior

- $q(x_{t-1} \mid x_t, x_0) = N\big(\bar{\mu}_t(x_t, x_0), \bar{\beta}_t I\big)$
- $\bar{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0 + \frac{\sqrt{1-\beta_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t$
- $\bar{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$
- Conditioning trick: the secret sauce that makes training tractable

The True Posterior — illustration

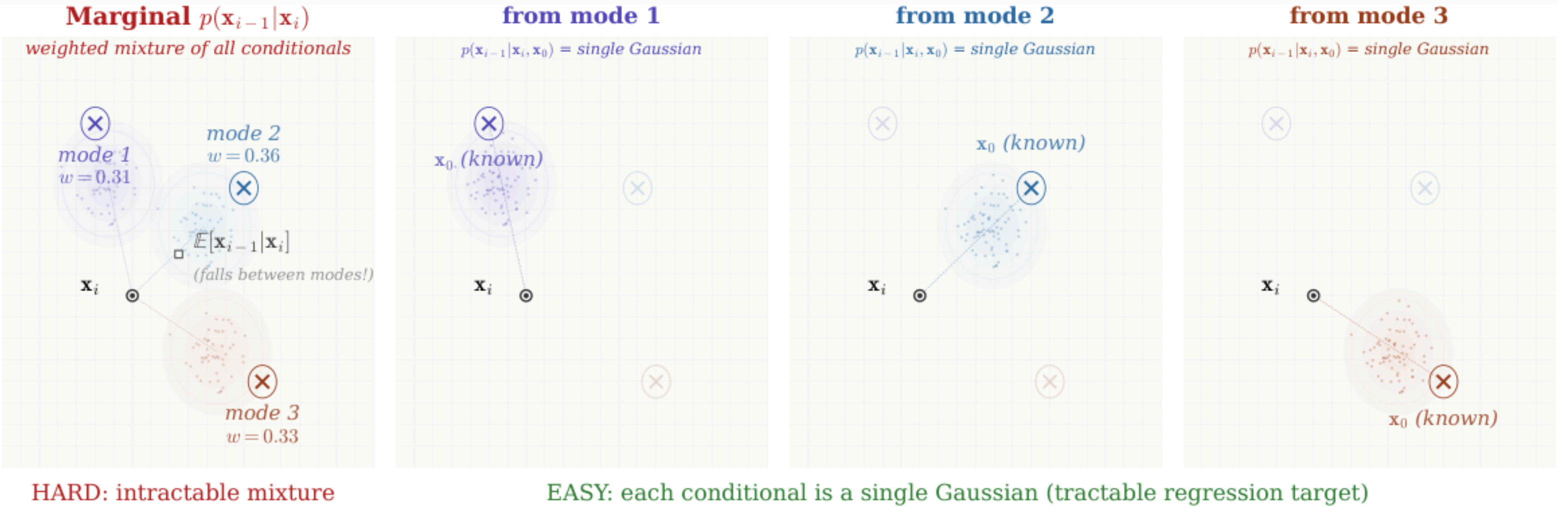


Figure 2.6: Illustration of the conditioning trick in DDPM. For a fixed noisy sample $\mathbf{x}_i$, the true reverse distribution $p(\mathbf{x}_{i-1}|\mathbf{x}_i)$ is induced by marginalizing over all possible clean origins $\mathbf{x}_0$, and is therefore generally a mixture that can be multimodal. Its mean may even fall between modes, making direct prediction difficult. By contrast, once we condition on the clean sample $\mathbf{x}_0$, the reverse posterior $p(\mathbf{x}_{i-1}|\mathbf{x}_i,\mathbf{x}_0)$ becomes a single Gaussian. DDPM exploits this tractable conditional structure during training, replacing a difficult multimodal prediction problem with a tractable regression target.

Parameterizing the Reverse Step

- Match $p_\theta(x_{t-1}|x_t) = N\big(\mu_\theta(x_t, t), \bar{\beta}_t I\big)$ to the posterior
- Predict x_0, the noise ε, or the velocity v — equivalent
- ε-prediction: μ_θ written via $\varepsilon_\theta(x_t, t)$
- Network sees (x_t, t) only — never the true x_0

Training DDPM

Variational Bound to Simple Loss

- ELBO over the chain = $L_T + \sum_t L_{t-1} + L_0$
- Each $L_{t-1} = D_{\mathrm{KL}(q(x_{t-1}|x_t,x_0)\,\|\,p_\theta(x_{t-1}|x_t))}$
- KL of two Gaussians = weighted $\|\bar{\mu}_t - \mu_\theta\|^2$
- Substitute ε-parameterization $\rightarrow$ clean noise-prediction loss

The ε-Prediction Objective

- $L_{\text{simple}} = \mathbb{E}_{t,x_0,\varepsilon}\left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2\right]$
- $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon$
- Drops the awkward per-t weights — works better in practice
- Tweedie links ε to the score: $\nabla \log p_t(x_t) = -\frac{\varepsilon_\theta}{\sigma_t}$ (Day 7)

Sampling: Denoise then Re-noise

- From x_t: predict ε_θ, estimate $\hat{x}_0$, jump to posterior mean
- Add a little fresh noise (except at the last step)
- Repeat $T \to 0$ — this is ancestral sampling
- Day 8: do this in far fewer steps with ODE/SDE solvers

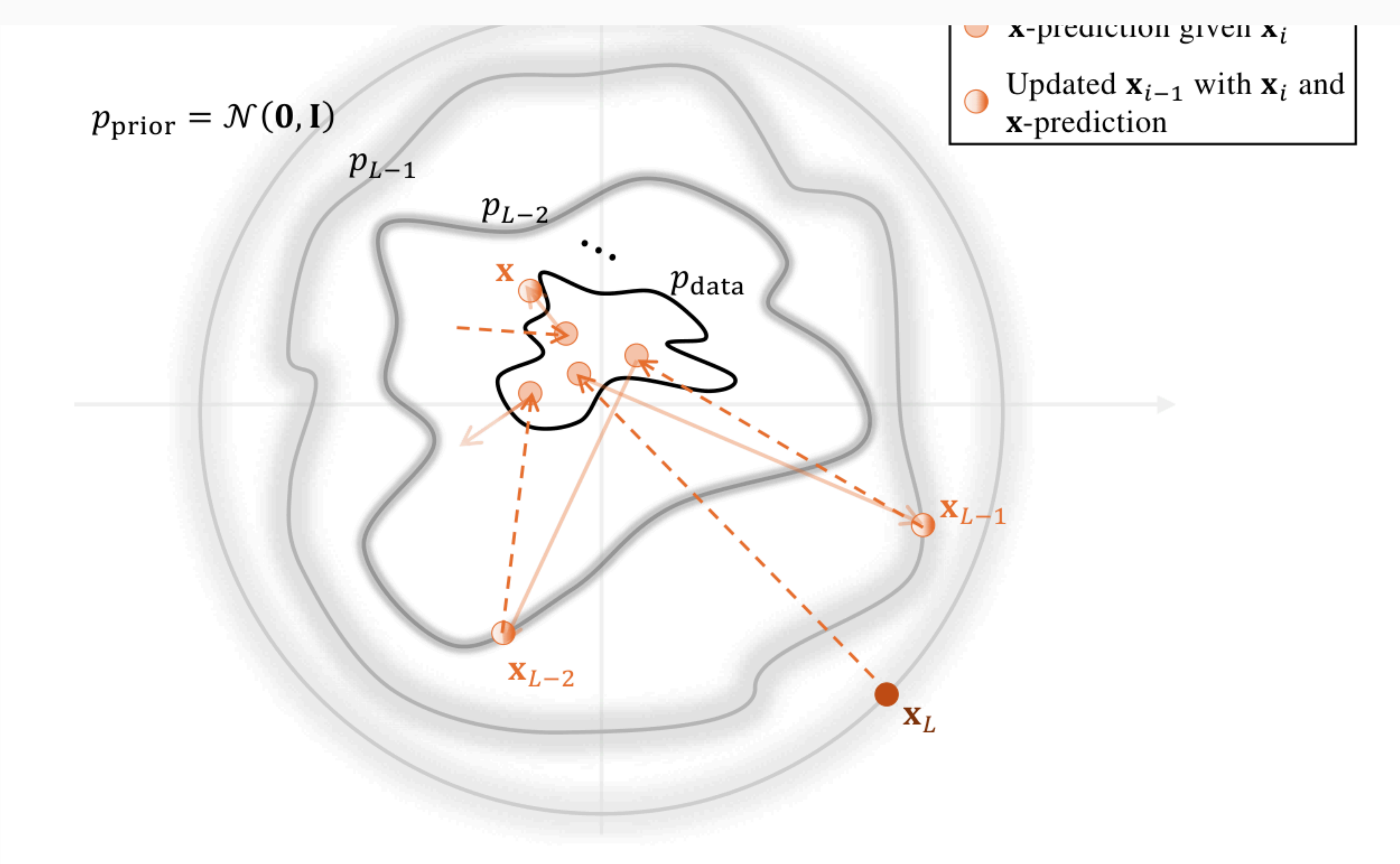


Figure 2.7: Illustration of the denoise-then-re-noise view of DDPM sampling. From the current

Sampling: Denoise then Re-noise — illustration

Training DDPM — Sampling: Denoise then Re-noise (source: course materials)

Summary

- Day 6: **Generative Modeling & DDPM**
- The variational view — from VAEs to diffusion
- Questions welcome — practical follows

Questions?

Practical session follows — see course site.